

Exploder Documentation

Version 1.3.7



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1 Introduction

Exploder is a Unity3d plugin that allows you to explode any GameObject that contains a mesh in **real-time**.

The destruction calculation is processed in **real time**. There are no pre-calculations or predefined GameObjects. Just put your beloved GameObject in the scene, Tag it with "Exploder" and watch the explosion!



Destroying vases from the Demo

2 How does it work?

The library contains a powerful **mesh cutter** that finds a mesh in your GameObject. After that it cuts the mesh recursively to small pieces, assigns a rigid body and velocity to each of them and makes an explosion.

For best possible performance the fragments are pre-allocated in a **pool** to minimize creating and instantiating GameObjects.

Cutting algorithm is very fast however you can specify **time budget** which is the maximum time to spend on calculation in a single frame. This allows you to create a powerful explosion in few frames with **low or no FPS drop**.

3 Quick start

The usage is **very simple**:

1. Create empty GameObject and rename it to "*Exploder*"
2. Assign ExploderObject component to "*Exploder*"
3. Adjust parameters in the inspector (or from the code)
 - o radius
 - o target fragments
 - o ...
4. Tag all objects you want to explode with the name "Exploder" (if you can't see "Exploder" in tag list, you have to create one)

5. Move “Exploder” GameObject to desired position so it covers your objects with the red sphere radius.
6. Create a demo script with public variable to access “Exploder” and call Explode(...)
7. Watch the explosion!

Tip: you can use Exploder prefab located in:

Exploder/Prefabs/Exploder.prefab

For the script reference please see the attached demo examples in [DemoClickExplode.cs](#).

This will guide you how to create a simple scene (also included in the package). Please watch the Youtube video on this address:

<http://youtu.be/u4WCLB6quBA>

4 How to use it in my FPS game?

Tip: from version 1.3.2 you can integrate Exploder with UFPS camera project, see more information in [this chapter](#).

This package also contains a first person shooter demo to demonstrate how the Exploder can be used with various weapons.

Shotgun example from the Demo

The basic idea:

1. Create empty GameObject with ExploderObject component.
2. (Setup your weapon and camera) and shoot with your mouse.
3. At the same moment run a raycast in direction from the camera.
4. Get an intersecting GameObject (ex. Vase).
5. Move ExploderObject to the position of the intersecting GameObject (ex. now the ExploderObject has the same position as the Vase).
6. Call Explode()
7. Watch explosion!

RPG example from the Demo

The setup for the RPG or Grenade is very similar. Just shoot a rocket (or throw a grenade) and when it hits the object (or after a timeout) move your ExploderObject to the same position and call Explode().

You don't need to maintain several ExploderObject components. Just only one and always move it to desired position.

All **source code is included** so you can see how the demo is constructed.

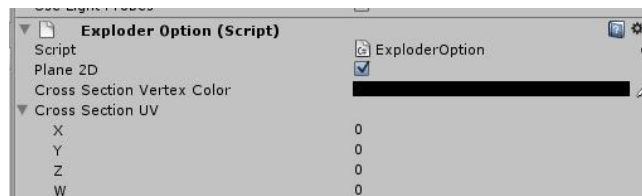
5 Exploding 2D sprite (Plane mesh)

Tested with 2D Toolkit.

(For the 2D physics support, see the [2D Physics chapter](#))

Exploder also works very well with 2D sprites or plane meshes. The only thing you have to do is to mark the object in **ExploderOption** settings.

ExploderOption is another script object you can find in the asset folder together with ExploderObject. You have to **assign this script** to your game object (sprite) you want to explode. There are several options in the script:



As you can see there is a checkbox "*Plane 2D*". By selecting it you specify that this object (owner game object) is a 2d plane or a 2d sprite. Later when you run the explosion Exploder will look for the ExploderOption and if it is marked as 2d plane it will change the explosion calculation to work with 2d meshes.

In short:

1. Find ExploderOption script and assign it to your 2d game object
 2. Check the "*Plane 2D*" option
 3. Run the explosion!
-

6 Cracking object (pre-calculated explosion)

Cracking the game object is another feature that allows you to run explosion calculation = prepare the object for explosion and later immediately execute the explosion.

The **advantage** of using the *crack* is **PERFORMANCE**. You can simply run the expensive calculation a long time before the explosion and then execute the explosion instantly. This can be useful especially on mobile devices, where the performance is lower than on desktop machine.

Usage:

Usage is similar to calling Explode(), you only need to call Crack(...) for cracking the object and later call ExplodeCracked(...). For script reference please see the attached example:

[DemoClickExplode.cs](#)

You only need to uncomment the macro ENABLE_CRACK_AND_EXPLODE on the third line to see cracking in action.

The **disadvantage** of using crack is that you cannot use more than one cracked object at the same time. You can crack only one object at the time, explode it and then crack a new one.

7 Using multiple Exploders at the same time

Using multiple Exploder objects at the same time to achieve simultaneous explosions is **NOT SUPPORTED** and **NOT RECOMMENDED**.

The reason why is very simple. Exploder calculation can be very CPU expensive, during calculation it takes as much time as specified in FrameBudget parameter. Imagine this scenario with two Exploders:

- Exploder_1 with FrameBudget = 15 [ms]
- Exploder_2 with FrameBudget = 15 [ms]
- Game is running in average 30 FPS, Exploder_1 runs explosion and takes approximately half of the framerate, the game experience is still acceptable. Exploder_2 runs another explosion at the same time and takes approximately another half of the framerate. The game **FPS reaches 0 and player is experiencing lag** that can take several frames.

If you really need to run as much explosions as possible in small amount of time, you can always increase the value of FrameBudget. For example if you set FrameBudget = 100, you can easily run more than one explosion in one or two frames, however it can heavily impact your framerate.

8 Custom texture for inner side of exploded fragments

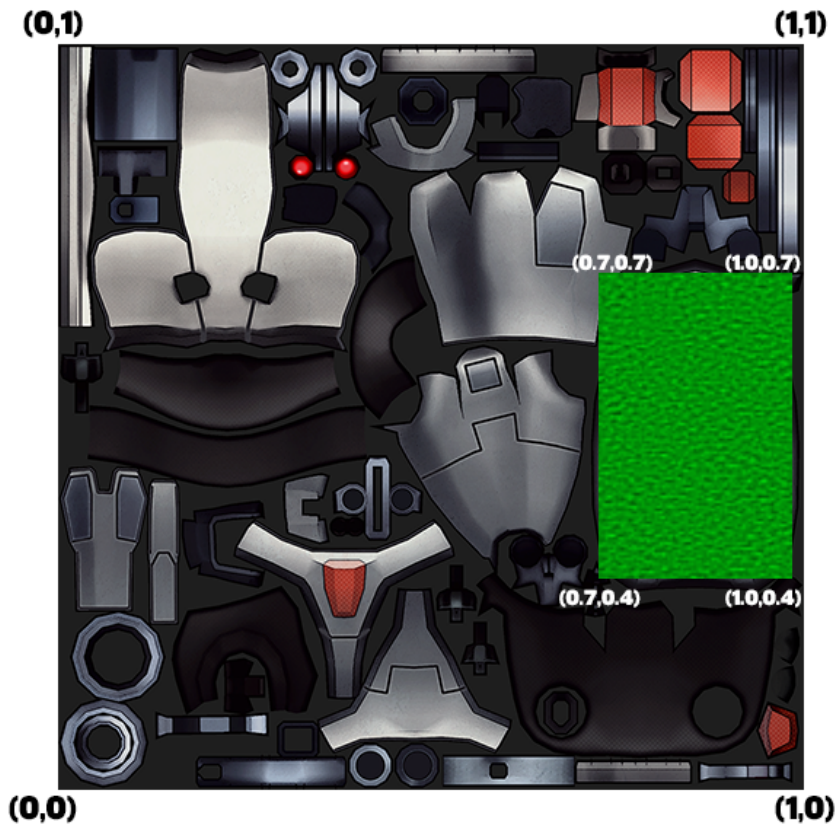
Specifying custom texture to the inner side of the fragment can be important for your game. For example exploding alien with green blood should show green texture inside the fragments. Exploder allows you to do that in slightly different way. You cannot use completely new texture for performance reasons because it would require creating lots of new submeshes on fragments. However you can specify UV portion of the original texture inside ExploderOption script:

Cross Section UV is the place where you can specify the UV mapping for the inner side of fragments.

- X is a starting **U** coordinate (between 0..1)
- Y is a **V** coordinate (between 0..1)
- Z is a ending **U** coordinate (between 0..1)
- W is a ending **V** coordinate (between 0..1)

Example with green blood alien:

This texture is an example how the character texture can look like. As you can see the UV coordinates are in range between 0 and 1. The green rectangle represents the *alien blood*.



To use the green texture you need to specify the coordinates in ExploderOptions like this:

9 Deactivation options

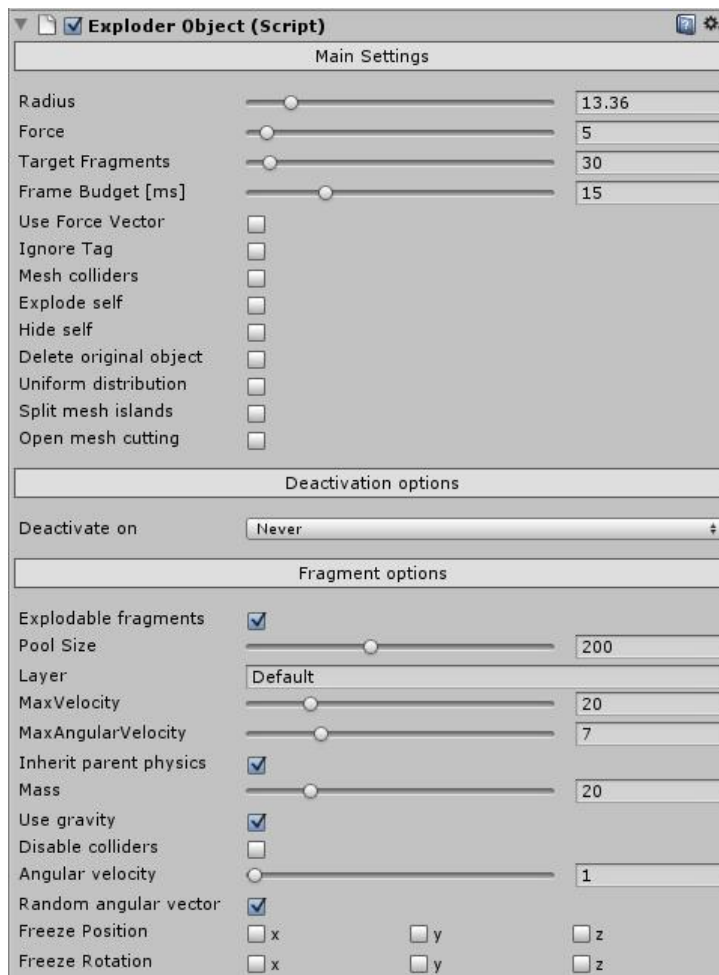
Deactivation options allows you to specify when the fragment can be deactivated.

- **Never** means fragment will be active all the time
- **OutsideOfCamera** means fragment will be deactivated once the main camera doesn't see it
- **Timeout** - means fragment will be deactivated after specified time [s] after explosion

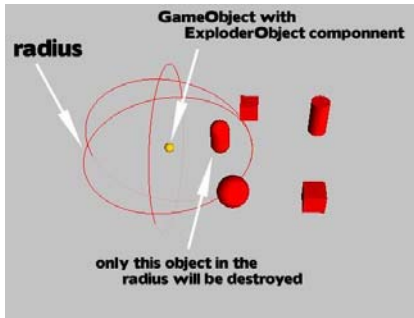
In case of **Timeout** deactivation, you can also choose whether you want to fade-out the fragments:

- **None** - no fade-out option
 - **FadeOut Alpha** - fragments will smoothly change its material alpha to zero and they will become fully transparent
 - **ScaleDown** - fragments will be smoothly scaled down to (0, 0, 0)
-

10 ExploderObject settings



Main Settings

Ignore Tag	Flag for not tagging Explodable objects. If you set this to TRUE you will have to assign "Explodable" script object to your GameObject instead of Tagging it. This is very useful if you already have tagged GameObject and you don't want to re-tag it to "Exploder".
Radius	Radius of explosion is a radius range in which the Exploder will be looking for destroyable objects. Notice the Red wire-frame gizmo in scene view. 
Target Fragments	Number of fragments that will be created by cutting the exploding objects. More fragments means more calculation and more PhysX overhead and also better looking explosion.
Frame Budget	Time budget in [ms] for processing explosion calculation in one frame. If the calculation takes more time it is stopped and resumed in next frame. Recommended settings: 15 - 30 (30 frames-per-second takes approximately 33ms in one frame). However this settings depends on your game and average frames-per-second.
Force Vector (and Use Force	Note: this option is valid only if UseForceVector is true. ForceVector represents a 3d Vector direction in which the explosion fragments will be moving. For example with

Vector)	Vector(0, 0, 1) exploding fragments will fly in "UP" direction. In the Demo Shotgun weapon is using ForceVector in direction of the camera, so the fragments are always moving from the player.
Mesh Colliders	Using mesh colliders for fragments. Use this option wisely because mesh colliders are very slow and you should avoid them in most cases. Mesh colliders can be very useful if you want to run explosion with small or zero force.
Force	Force is how much the physical force is added to exploding fragments. More force means higher velocity.
Explode Self	Flag for destroying mesh in owner GameObject if there is any mesh component in it.
Disable radius	This option is valid only if ExplodeSelf is set to true. Disable radius scan for object, only the parent object will explode and nothing else.
Hide Self	Flag for hiding owner game object after explosion. This can be useful if you want to only hide and not to destroy owner object.
Destroy Original Object	Flag for destroying original game object after explosion. For example when you destroy a "chair" model, it will cut the chair to small pieces and create new fragment game objects. If this flag is true it will call Object.Destroy(chair) and destroy the original chair GameObject from the game. Otherwise the original object (chair) will be only deactivated.
Split mesh islands	Option for separating not-connecting parts of the same mesh. If this option is enabled all exploding fragments are searched for not connecting parts of the same mesh and these parts are separated into new fragments.
Uniform distribution	By enabling this Exploder will create number of fragments per object equally regardless of object distance from the center. By default objects closer to the center (exploder center) will be shattered into more fragments than objects far from the center. Uniform distribution will guarantee that all objects will be shattered into same number of fragments.
Open Mesh Cutting	This option will allow you to use exploder with non-closed meshes, for example airplane with windows (mesh with holes). However it will not always triangulate the fragments properly. Use this only if your object has holes inside and/or it doesn't want to explode.

Fragment options

Explode Fragments	Flag for destroying already destroyed fragments. If this is true you can destroy the object and then all its fragment pieces. You can keep destroying fragments until they are small enough.
Fragment Pool Size	Maximum number of all available fragments. This number should be higher than Target Fragments.
Max Velocity	Maximum velocity for fragments, higher velocity will be clamped.
Max Angular Velocity	Maximum angular velocity for fragments, higher velocity will be clamped.
Random Angular Vector	This will randomize fragment rotation after explosion. If this item is false you will have specify axis of rotation manually.
Angular Velocity	Angular velocity for fragments, if "Inherit parent physics" is enabled, final angular velocity will be calculated as a sum of parent and this value.
Disable colliders	Disable colliders from all fragments, use this if you don't want the fragments collide.
Inherit parent physics	By enabling this fragment will use same physics properties as its parent rigid body. It will inherit Mass, Velocity, Angular Velocity and Use Gravity. If there is no valid parent rigid body default settings will be used instead.
Mass	Mass of the fragment. if the parent object has rigidbody and Inherit Parent Physics is true the mass property for fragments will be calculated

	based on this equation ($\text{fragmentMass} = \text{parentMass} / \text{TargetFragments}$)
Layer	Name of layer for fragments. Use this if you want to change default layer for fragments.
Freeze position	Freeze position of fragment in specified axis.
Freeze rotation	Freeze rotation of fragment in specified axis.
Material	Optional material for fragments, if not selected default material (first) from the original object is used

SFX options

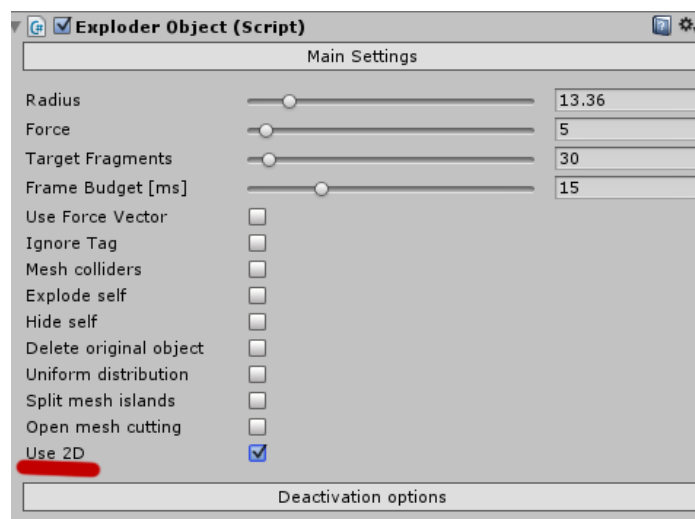
Explosion sound	Optional AudioClip that will be played on the beginning of explosion.
Fragment hit sound	Optional AudioClip that will be played when one of the fragment piece hit another object (collider).
Hit sound timeout	This option is valid only if Fragment hit sound AudioClip is set. Timeout between AudioClips, this is useful to prevent over-halting audio system when the explosion has lots of fragments.
Fragment particles	GameObject with particle emitters, that will be run when the explosion starts. Particles will be emitted from every fragment.
Maximum emitters	This option is valid only if Fragment particles is set. Maximum number of particle emitters, use low number if your game is experiencing lag.

11 2D Physics (Unity4.3+)

Tested with 2D Toolkit.

Exploder support 2D physics as well. You can enable this feature by setting Use2Dcollision (Use 2D) to True. All explosion fragments will have 2D colliders (PolygonCollider2D).

It is important that you will not use this settings with exploding 3D objects (it might work but it might lead to unexpected behavior).



<http://youtu.be/IZWz5mt2kIY>

IMPORTANT NOTE:

Exploder DOES NOT work with Unity4.3+ built-in sprites, because there is no way to access sprite mesh data required by Exploder.

12 UFPS integration (version 1.3.7)

Exploder has been successfully tested with Ultimate FPS camera 1.4.6, all you need to do is to copy/paste few script methods as described below.

1. Add Exploder prefab to the scene (Exploder/Prefabs/Exploder.prefab). It is important to use this prefab because it also contains ExploderSingleton script for easy access of Exploder object.
2. Open file vp_HitscanBullet.cs in UFPS folder.
3. Add these two methods to the vp_HitscanBullet class:

```
private void ApplyExploder(GameObject target)
{
    var exploder = Exploder.Utills.ExploderSingleton.ExploderInstance;
    if (exploder)
    {
        // check if the target is explodable object
        if (Exploder.ExploderUtills.IsExplodable(target))
        {
            // activate exploder
            Exploder.ExploderUtills.SetActive(exploder.gameObject, true);

            // move exploder object to the same position
            exploder.transform.position
= Exploder.ExploderUtills.GetCentroid(target);

            // decrease the radius so the exploder is not interfering with other
objects
            exploder.Radius = 1.0f;

            // DONE!
            exploder.Explode(OnExplosion);
        }
    }
}

private void OnExplosion(float time, Exploder.ExploderObject.ExpllosionState state)
{
    if (state == Exploder.ExploderObject.ExpllosionState.ExpllosionFinished)
    {
        //Utills.Log("Exploded");
    }
}
```

4. Locate method `void DoHit()` in the same class and find this code, it should be around line 158:

```
hit.collider.SendMessageUpwards(DamageMethodName, Damage, SendMessageOptions.DontRequireReceiver);
```

5. After this line add this command:

```
ApplyExploder(hit.collider.gameObject);
```

6. Done! You can now fine tune Exploder settings and see explosions in the game.

Tip: if you have a diff tool, you can use a patch located in:

Exploder/UFPS/patchUFPS_1.4.6.patch

13 Playmaker support

There are 3 Exploder custom actions ready to work with Playmaker:

- Crack
- Explode
- ExplodeCracked

To enable these actions you have to open the action scripts located in Assets/Exploder/Playmaker/* ...and uncomment the second line (//#define PLAYMAKER):

from this:

```
// uncomment next line to work with Playmaker
//#define PLAYMAKER
#if PLAYMAKER
```

to this:

```
// uncomment next line to work with Playmaker
#define PLAYMAKER
#if PLAYMAKER
```

14 FAQ

Q: I cannot explode my objects. Sometimes it creates only one big fragment.

A: Make sure your objects are in exploder radius, properly tagged and the TargetFragments value is high enough. If it still does not work you can try to enable "Open mesh cutting" which is an option for exploding non-closed meshes. This option makes Exploder more tolerant to complex meshes with holes. If it still does not work, please contact me by e-mail with screenshot of your object hierarchy and your model in scene.

Q: Explosion seems to be huge even if I set small Force value.

A: It is because all fragments have by default box colliders and they are tight together overlapping each other, which might cause high explosion regardless of Force value. You can easily fix it by enabling **Mesh colliders** and/or limiting velocity with **MaxVelocity** parameter.

Q: FadeOut Alpha does not seem to work. Fragments disappear in one frame (blink).

A: FadeOut Alpha works only with transparent materials that allow to change alpha value.

Q: Can I specify custom texture for fragments?

A: You can specify UV portion from parent texture, see more details in this [chapter](#).

Q: Does Exploder support reverse explosion?

A: Unfortunately not, Exploder can only explode objects in usual way.

Q: Can Exploder destroy Unity sprite?

A: Unity built-in sprites are not supported by Exploder because there is no way to access mesh data, this feature will be hopefully available in future version of Unity.

Q: I don't want to change the tag to "Exploder", it already has one.

A: You don't have to, just make sure to enable "Ignore Tag" and add "Explodable" component to your game object.

Q: Can Exploder destroy only a portion of the object, like a head or arm of a character?

A: Unfortunately it can't, Exploder works only with a whole mesh, in case of a character it will explode head, arm, body, legs and everything else at once.

Q: Does Exploder support submeshes?

A: It does, but in limited way. It can explode mesh with submeshes but resulting fragments will have only one mesh. It means that all fragments will have only one material.

Q: How to access fragment pool and get active fragments?

A: Please refer to file **HowToGetActiveFragments.cs**.

Q: Can I use Exploder to create fragment prefabs?

A: No, Exploder is only for real-time use. However you can use Crack() feature to pre-calculate explosion and execute it later.

Q: Any tips for increasing performance?

A: First of all **TargetFragments**, try to keep it low, less fragments mean less overhead and less calculations. Also make sure that **FrameBudget** is lower than half of your expected frame rate, for example game with 30 FPS should have **FrameBudget** 15 and less, otherwise you can notice lagging. Try to **avoid Mesh colliders**, they can be very expensive. Also don't use **SplitMeshIslands** so much, it can save you few more frames. Don't forget that performance is always about tradeoff, good luck with your game!

15 Support

If you have any questions or need a help with setting up the Exploder game object you can write to unity forum:

<http://forum.unity3d.com/threads/190198-Exploder-RELEASED>

or you can send me an e-mail to:

gamesreindeer@gmail.com